

DMA LAB-7 R BASICS

LOADING THE DATA

```
> df <- read.csv("D:/fruits.csv")
> head(df)
  weight size color_score Fruit_label
1    150  7.5         0.50        lemon
2    170  7.5         0.50        lemon
3    190  8.0         0.74        orange
4    210  8.5         0.75        orange
5    230  8.5         0.75        orange
6    250  8.5         0.80         Apple
```

DESCRIPTION OF DATA

```
> str(df)
'data.frame':  23 obs. of  4 variables:
 $ weight      : int   150 170 190 210 230 250 270 290 165 185 ...
 $ size        : num   7.5 7.5 8 8.5 8.5 8.5 8.5 9 7.5 7.5 ...
 $ color_score : num   0.5 0.5 0.74 0.75 0.75 0.8 0.8 0.8 0.5 0.5 ...
 $ Fruit_label : chr   "lemon" "lemon" "orange" "orange" ...
```

CLASS DATATYPE

```
> class(df$color_score)
[1] "numeric"
```

LENGTH OF A CLASS

```
> length(df$color_score)
[1] 23
```

DIMENSIONS OF DATA

```
> dim(df)
[1] 23  4
> ncol(df)
[1] 4
```

CLASS NAMES / COLUMN NAMES

```
> names(df)
[1] "weight"      "size"        "color_score" "Fruit_label"
```

CENTRAL TENDENCY

```
> mean(df$weight)
[1] 231.3043
> median(df$weight)
[1] 230
```

MEASURES OF DISPERSION

```
> sd(df$weight)
[1] 45.95495
> var(df$weight)
[1] 2111.858
> cov(df$weight, df$size)
[1] 18.20158
> range(df$weight)
[1] 150 305
```

SUMMARY OF A CLASS

```
> summary(df$weight)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
150.0  195.0   230.0   231.3   267.5   305.0

> summary(df)
      weight      size    color_score  Fruit_label
Min.   :150.0 Min.   :7.500  Min.   :0.5000  Length:23
1st Qu.:195.0 1st Qu.:7.500  1st Qu.:0.5000  Class :character
Median :230.0 Median :8.000  Median :0.7600  Mode  :character
Mean   :231.3 Mean   :8.152  Mean   :0.7191
3rd Qu.:267.5 3rd Qu.:8.500  3rd Qu.:0.8100
Max.   :305.0 Max.   :9.000  Max.   :0.9200
```

TABLE

```
> table(df$weight>200)

FALSE  TRUE
   7    16

> table(df$Fruit_label=="Apple")

FALSE  TRUE
  16     7
```

IS NA TO FIND NULL VALUES

```
> is.na(df$weight)
 [1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[14] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

SUBSET OF A DATA

```
> df1=subset(df,Fruit_label=="Apple")
> df1
   weight size color_score Fruit_label
6     250  8.5         0.80      Apple
7     270  8.5         0.80      Apple
8     290  9.0         0.80      Apple
12    225  8.5         0.81      Apple
13    245  8.5         0.82      Apple
14    265  8.5         0.81      Apple
15    285  8.5         0.79      Apple

> df2=subset(df,weight>200)
> df2
   weight size color_score Fruit_label
4     210  8.5         0.75     orange
5     230  8.5         0.75     orange
6     250  8.5         0.80      Apple
7     270  8.5         0.80      Apple
8     290  9.0         0.80      Apple
11    205  8.0         0.75     orange
12    225  8.5         0.81      Apple
13    245  8.5         0.82      Apple
14    265  8.5         0.81      Apple
15    285  8.5         0.79      Apple
16    305  9.0         0.92 Pomegranate
19    220  8.0         0.91 Pomegranate
20    240  7.5         0.50      lemon
21    260  7.5         0.50      lemon
22    280  8.0         0.76     orange
23    300  9.0         0.91 Pomegranate
```

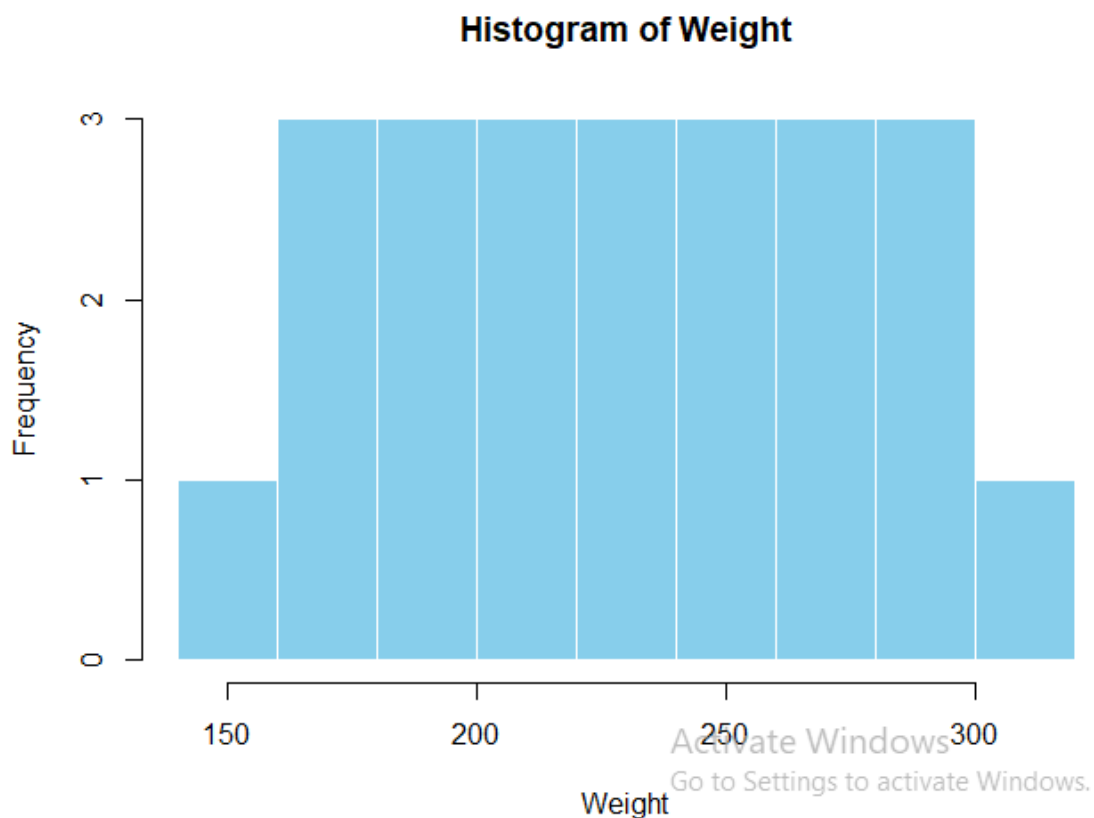
SUBSET WITH RANGE

```
> df3=subset(df,weight %in% c(200:300))
> df3
```

	weight	size	color_score	Fruit_label
4	210	8.5	0.75	orange
5	230	8.5	0.75	orange
6	250	8.5	0.80	Apple
7	270	8.5	0.80	Apple
8	290	9.0	0.80	Apple
11	205	8.0	0.75	orange
12	225	8.5	0.81	Apple
13	245	8.5	0.82	Apple
14	265	8.5	0.81	Apple
15	285	8.5	0.79	Apple
18	200	8.0	0.92	Pomegranate
19	220	8.0	0.91	Pomegranate
20	240	7.5	0.50	lemon
21	260	7.5	0.50	lemon
22	280	8.0	0.76	orange
23	300	9.0	0.91	Pomegranate

1. HISTOGRAM

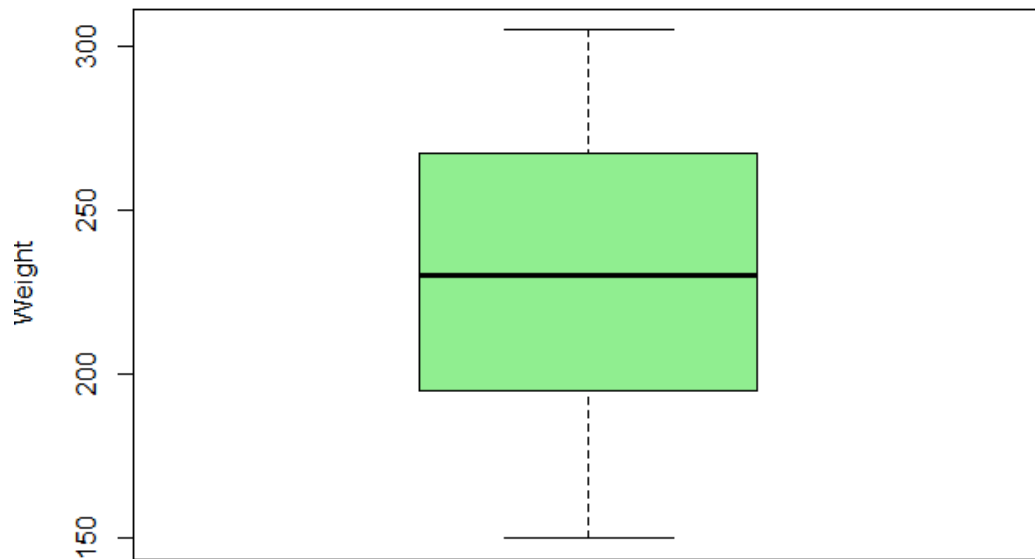
```
> hist(df$weight,
+      main = "Histogram of Weight",
+      xlab = "weight",
+      col = "skyblue",
+      border = "white")
```



2. BOXPLOT

```
> boxplot(df$weight,
+         main = "Boxplot of Weight",
+         ylab = "weight",
+         col = "lightgreen")
```

Boxplot of Weight



3. SCATTER PLOT

```
> plot(df$weight, df$size,  
+       main = "Scatterplot of weight vs Height",  
+       xlab = "weight",  
+       ylab = "Height",  
+       pch = 19,  
+       col = "blue")
```

Scatterplot of Weight vs Height

